

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Department of Geology and Geophysics

2. Subject Name: Borehole Geophysics Lab L-T-P: 0-0-3 Credits: 2

3. Pre-requisites:

4. Syllabus and reference books:

Syllabus:

1. Determination of Resistivity variations with temperature, salinity using charts.
2. SP log interpretations and determination of R_w from SP log.
3. Resistivity log interpretations and determination of R_w , S_w , S_{hy} using Archie's relation.
4. Induction log interpretations and determination of depth of invaded zones.
5. Porosity log interpretation using Neutron, Density and Sonic logs.
6. Quick look Approach of log Interpretations.
7. Shaly Sand formations interpretations using cross plots of Neutron and Density logs.
8. Complex Lithology interpretation using standard logs.
9. Hingle Plot interpretations.
10. Production log interpretations
11. Seismic to well ties in OpenDtect software.

Reference Books:

- 1)
- 2)
- 3)

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.		
2.		
3.		
4.		
...		
Total number of hours		

